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Intro to Python
Due: September 4th, 2026

Part 1: COBOL

1. Why does the author compare COBOL to asbestos? Do you think the comparison makes sense?

The author compares COBOL to asbestos because the program has inundated governmental and financial systems across the US and, much like asbestos, has proven arduous to remove in favor of contemporary programming languages. This comparison makes sense, as asbestos was widely adopted worldwide as a construction material, much like COBOL, the “preferred” programming language for the last 60 years. Many warned of the dangers of COBOL’s dominance, but these dissents were largely ignored.

2. What did the people who designed COBOL want it to be able to do? Do you think COBOL lives up to those expectations?

The designers of COBOL intended for the language to be easy to use for programmers and nonprogrammers alike. The designers built the language using plain English to ensure their primary objective, which contributed to COBOL’s commercial success. COBOL seemed to live up to those expectations; the true qualm people seem to have with COBOL is its lack of precision/wonky structure.

3. What do you think makes a programming language successful or unsuccessful? Is COBOL a successful programming language?

A programming language is only as successful as its accessibility. For example, the programming language Python has garnered commercial success that is directly tied to its ease of use and learning. In an ever changing world, the products and services that stand the test of time are those that fit into people’s everyday life and improve functioning. The more accessible a language is, the larger its market base/share becomes, thus garnering commercial success.

4. Do you think that the AI tools mentioned in the last paragraph of the article will be successful? Why or why not?

It seems that AI tools have proved largely unsuccessful in reversing the inundation of COBOL within systems. The author of the article, presumably more versed in the intricacies of programming than I, presents a convincing argument as to why AI tools currently aren’t enabled to tackle this problem. It seems that AI tools are not advanced enough to tackle problems such as the one the article poses of transitioning systems away from COBOL frameworks to more contemporary programming languages. The only solution evident to me is a complete rewrite of these systems in contemporary code.

Part 3: Producing Errors

1. The `TypeError` arose because you cannot use the multiplication operator to multiply a string and a list. These types are incompatible with the `**` operator.
2. This code raises a `SyntaxError` because it doesn't use parentheses when calling the built-in `print` function to display the string, which also is also missing its delimiters (e.g., `"` or `'`).
3. This code raises a `NameError` because the variable `y` is not defined, so its value cannot be printed. The first line also raises a `SyntaxError` because it attempts to assign a value to a variable using `is` instead of the assignment operator `=`.

Part 4: Definitions & Miscellany

1. How are binary and unary operators different? Name an example of each.

Binary operators take two operands. Some binary operators include: `+` and `-`. Unary operators take a single operand. An example is `-` which returns the inverse of a value.

2. Explain the difference between syntax and semantics as if you were talking to someone who is not familiar with programming.

Syntax is essentially having the right "ordering". It's how we arrange words and choose the correct forms of words to produce valid sentences. Semantics, however, refers to the "meaning". To verify whether a program is semantically sound, one should ask, "Does the code do what we intend for it to do?"

3. Explain the difference between static and dynamic typing as if you were talking to someone who is not familiar with programming.

With static typing, once we name a variable and assign it a value, it cannot be changed. For example, if we encode in our program that `x` is equal to 7, we cannot reassign `x` to be "Billy". The program will not run our code. However, in languages like Python that employ dynamic typing, we are allowed to reassign the value of a variable, and the code will run.

4. Explain the difference between an integer and a float as if you were talking to someone who is not familiar with programming.

An integer is a positive or negative whole number (e.g, 1, -2, 7, 95). Floats are numbers with decimal points.

Part 6: Write a Homework Problem

Design a function that converts military time hours in the set $\{1,24\} \in \mathbb{N}$ into standard time.

This homework question assesses the ability of students to apply modular arithmetic in real-life settings and practice using Python's function tool. This problem does not seem like a modular arithmetic problem on the surface, but when defining their function, students will need to utilize the skill to successfully design a function that converts military hours into standard time hours. Students may struggle with building the function itself. It seemed to be a skill that we all need to work on and did not feel confident with. In defining the program,s students may struggle with deciding between using floor division, integer and the correct remainder operands. In overcoming these difficulties, students will shore up their understanding of functions and the use of the aforementioned operands.

Note: This program is crude, but as we gain more competencies and skills, we will be able to improve this program.